Understanding gene regulation for genetic engineering

Understanding gene regulatory principles (Systems Biology) and applying gene regulatory principles for circuit design (Synthetic Biology)

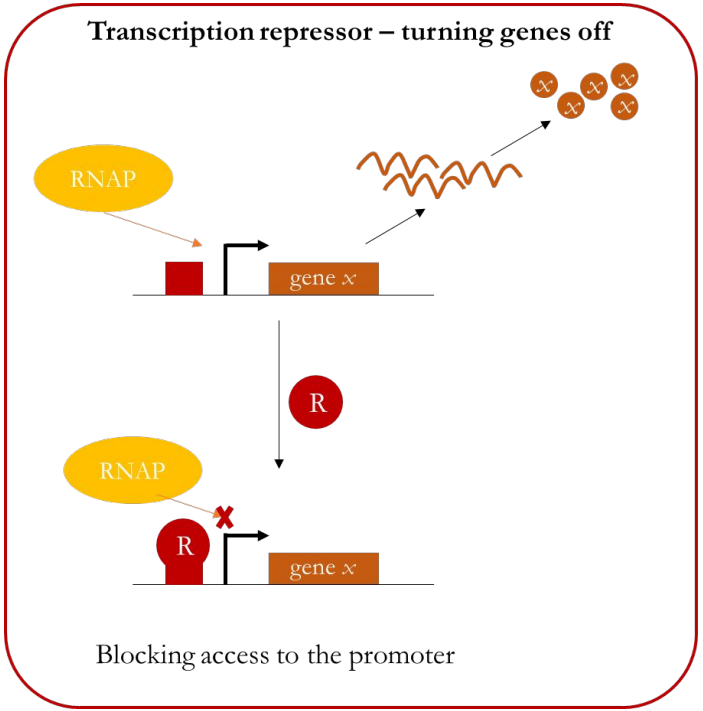
Lecture 2

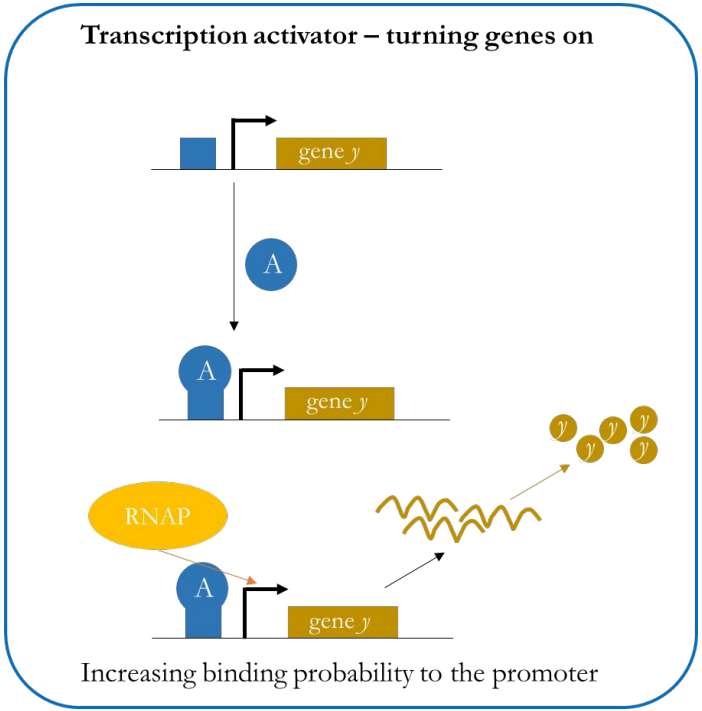
Somenath Bakshi

Laboratory of systems and Synthetic Microbiology

Information and Bio - engineering division, Department of Engineering

Transcriptional regulation – signs of regulation

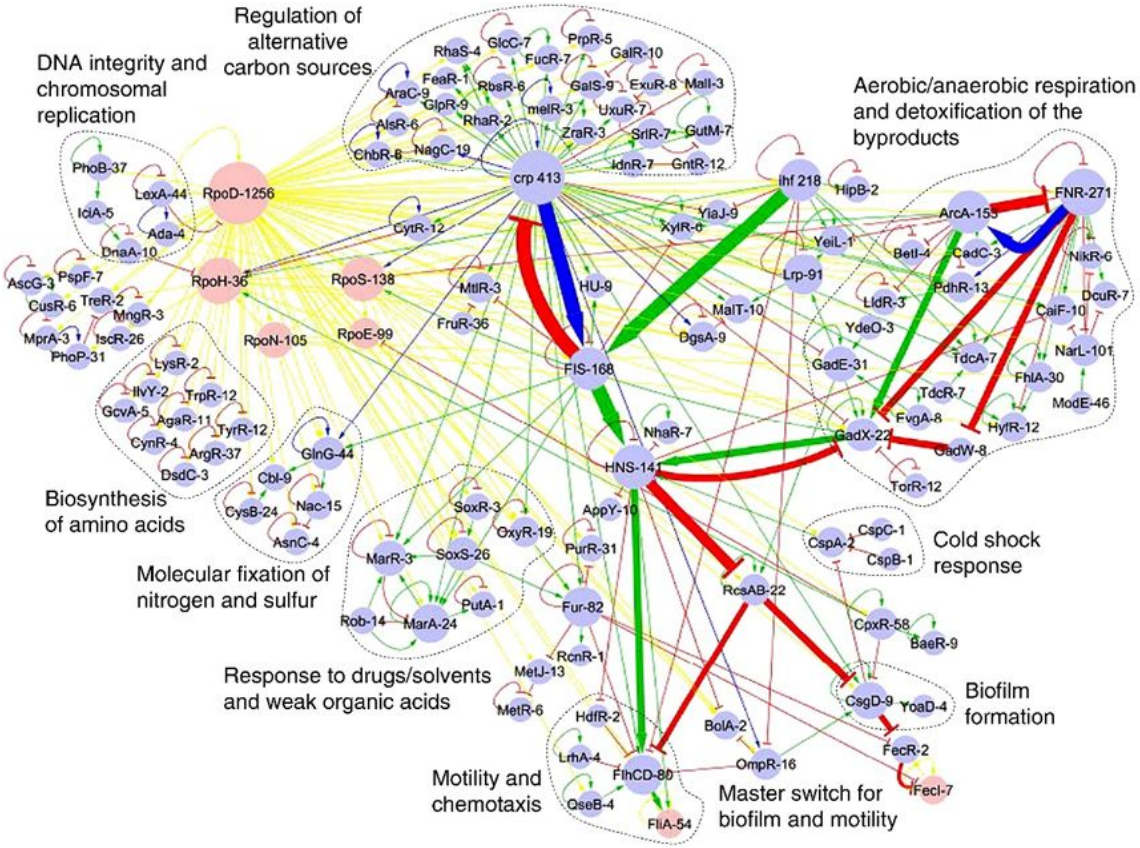




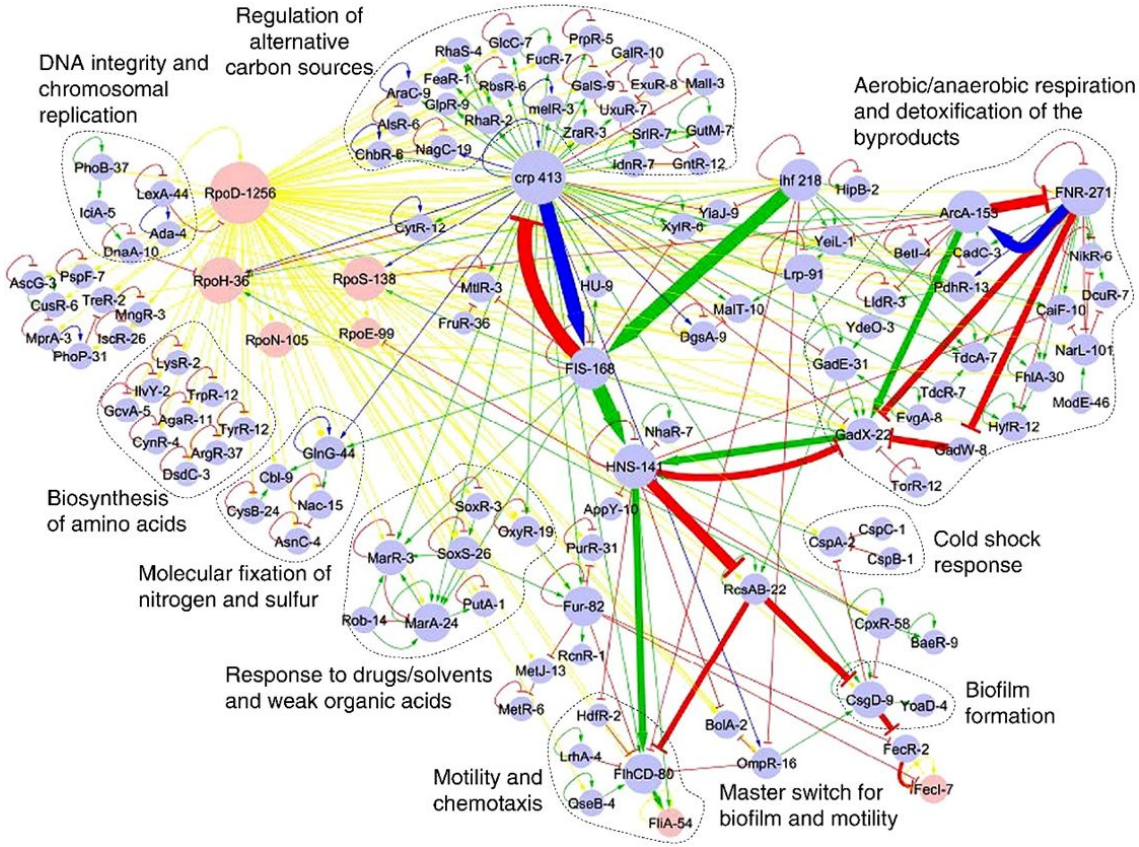




Gene-regulatory network of E. coli



Quiz 3

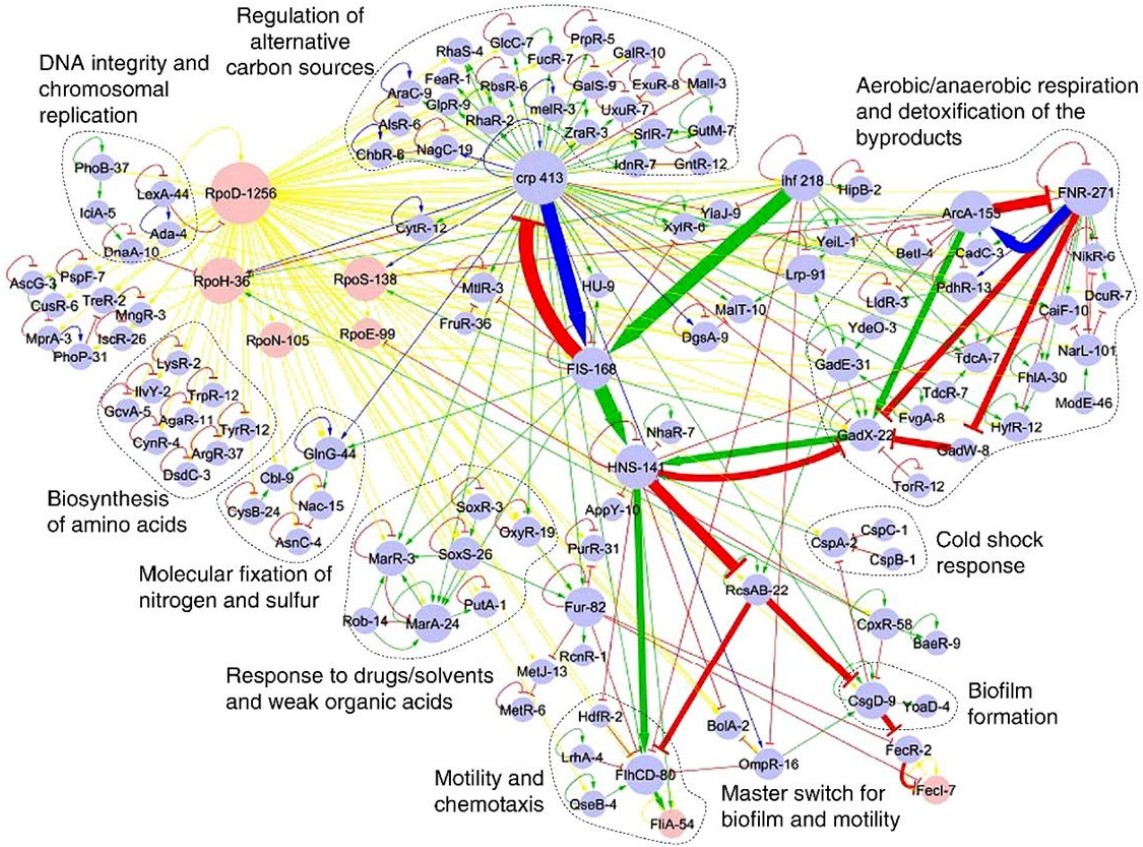


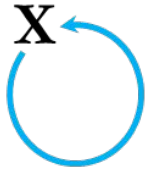
The gene regulatory network of *E. coli* has **424** nodes and **519** arrows.

It has **40** self arrows (**34** of these are repressors; **6** activators)

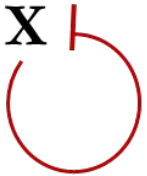
are self-arrows more frequent than random?

Self arrows in the gene-regulatory network of *E. coli*



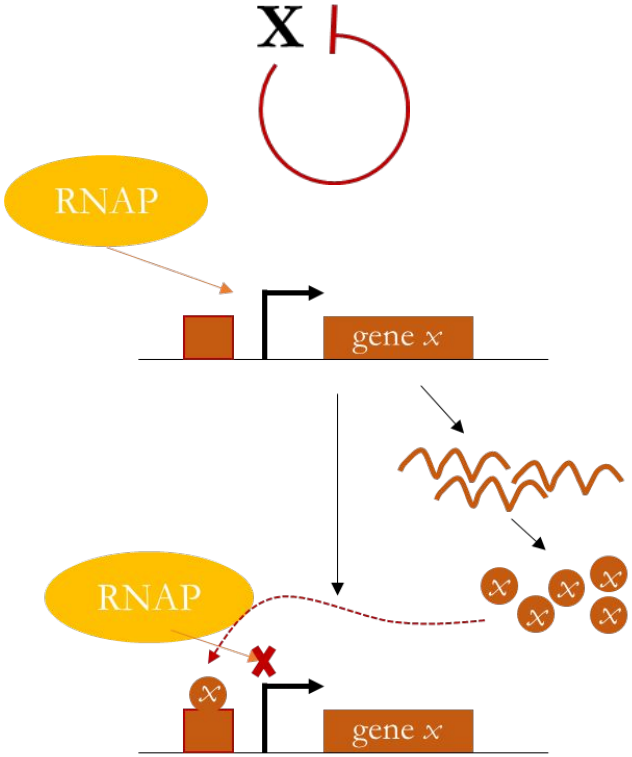


Positive autoregulation (PAR)



Negative autoregulation (NAR)

Negative autoregulation (NAR)

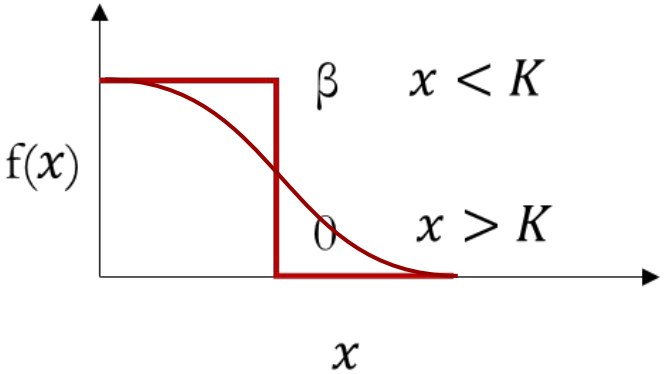


$$\frac{dx}{dt} = f(x) - \lambda \cdot x$$

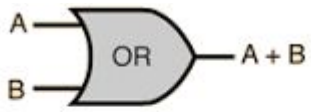
$$\text{Promoter activity} = \beta \cdot \frac{K^n}{K^n + R^n}$$ Hill function

Logic approximation for high n

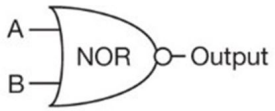
$$\text{Promoter activity} = \beta \cdot \theta \ (R < K)$$ → **Genetic logic gates**



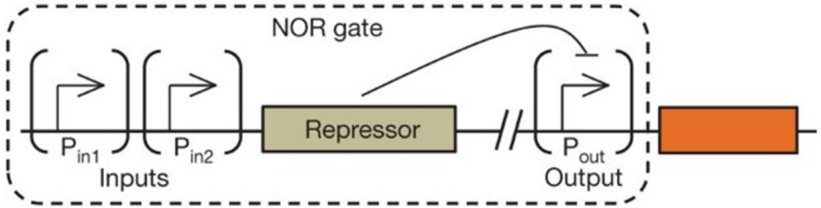
Genetic logic gates: OR and NOR

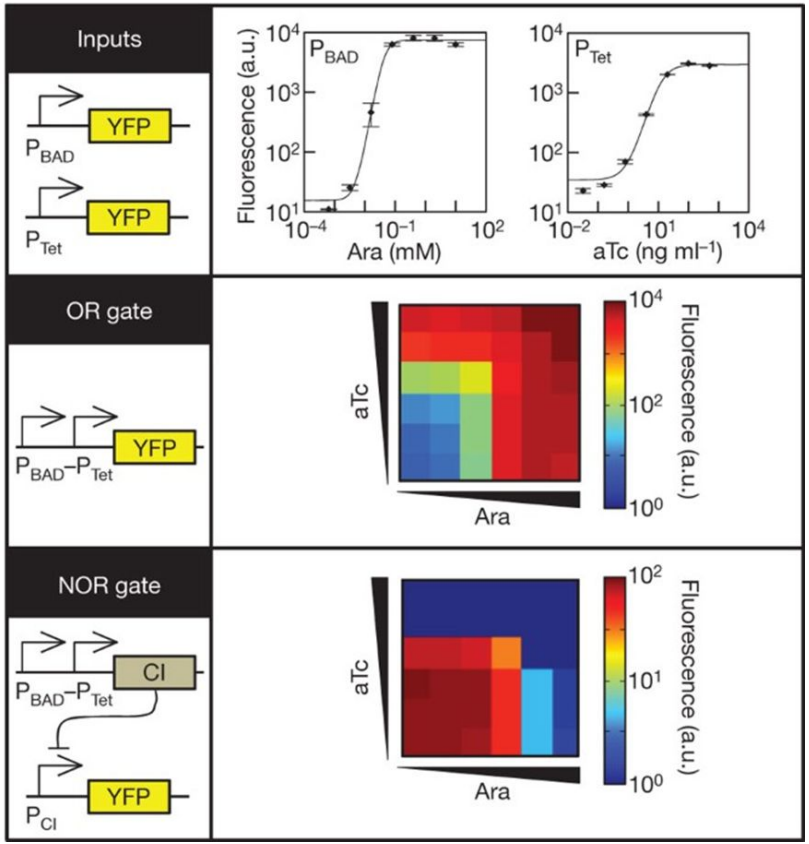


A	B	Out
0	0	0
0	1	1
1	0	1
1	1	1



Inputs		
in1	in2	Output
0	0	1
0	1	0
1	0	0
1	1	0

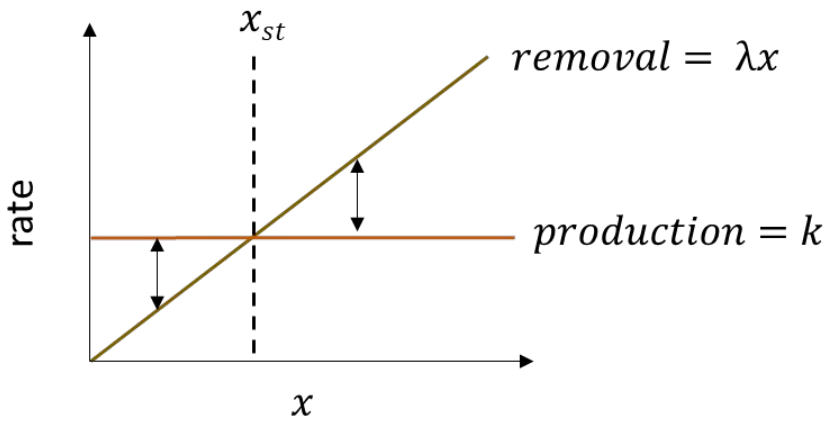




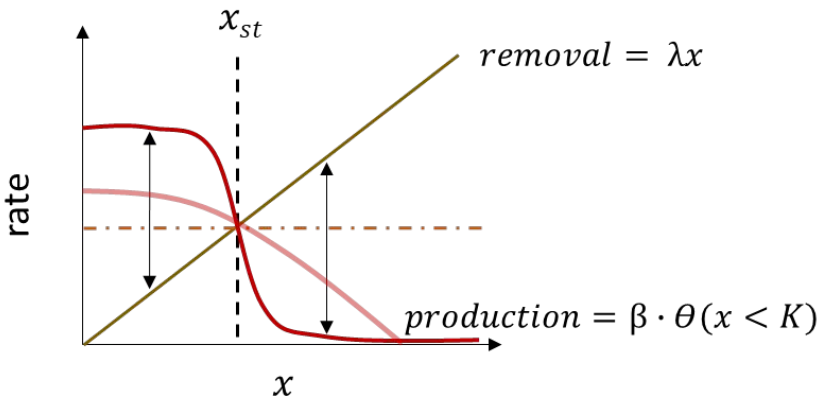
PMID: 21150903

NAR enables fast response: rate plots

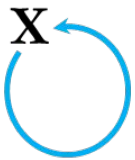
Simple regulation

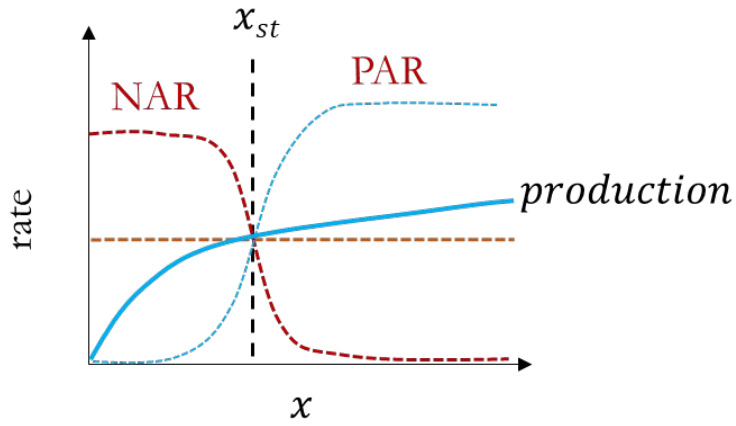


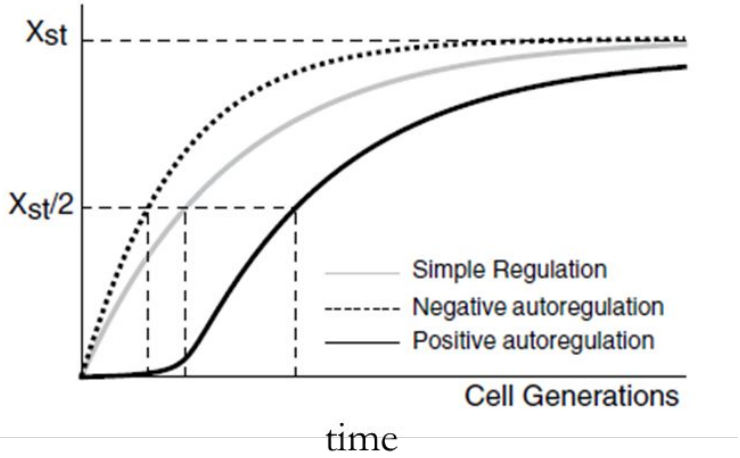
Negative autoregulation



Positive autoregulation (PAR)



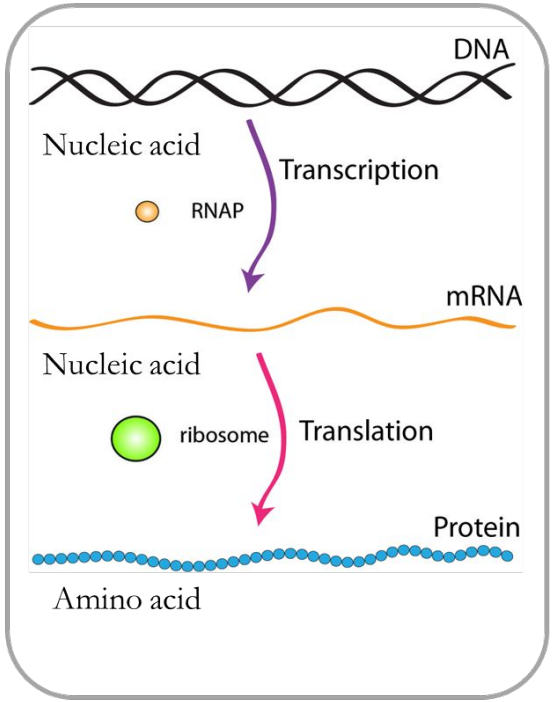




What is the usefulness of positive autoregulation?

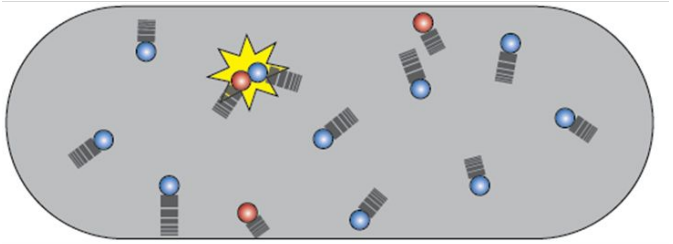
Stochastic gene-expression

Steps in gene-expression



Central dogma of molecular biology

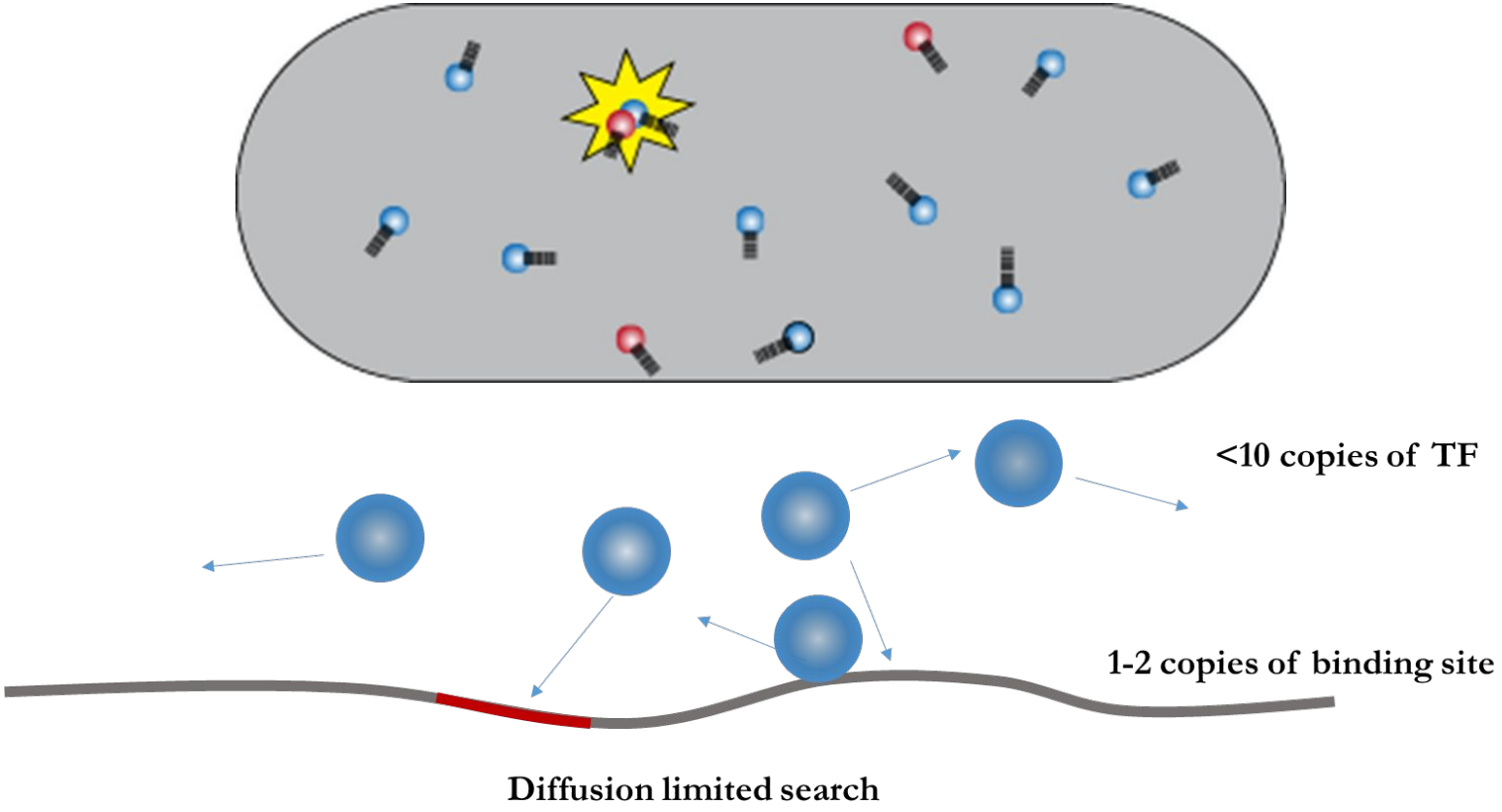
Species	Copy number	
Active gene	0-5	
mRNA	0-25	
Protein	0-250	**(TFs)**



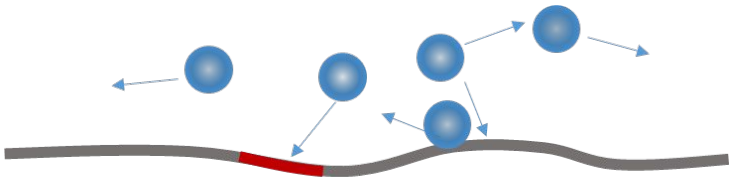
Individual chemical reactions in cells are probabilistic

(PMID: 33835126)

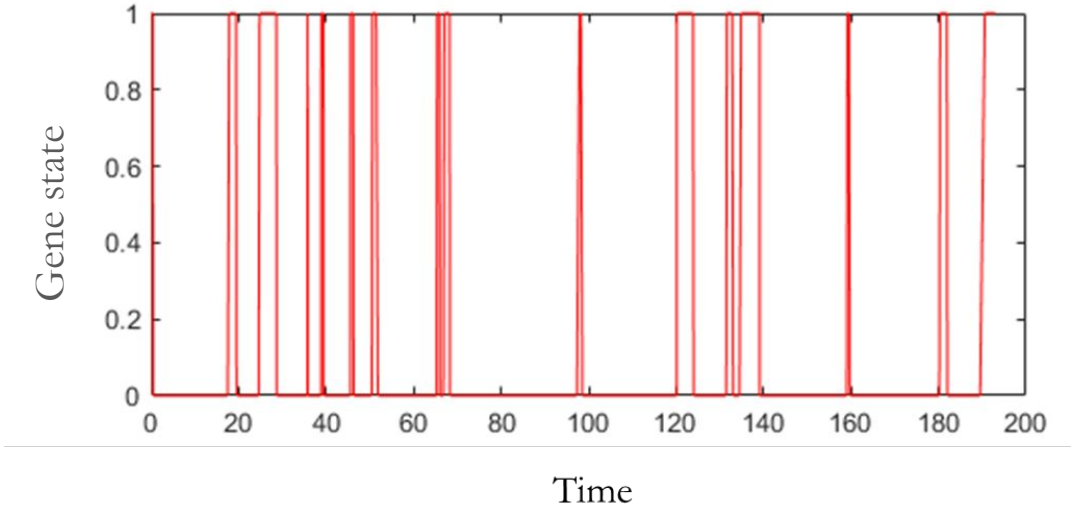
Random collision between low-copy players: gene activation



Stochastic activation of genes

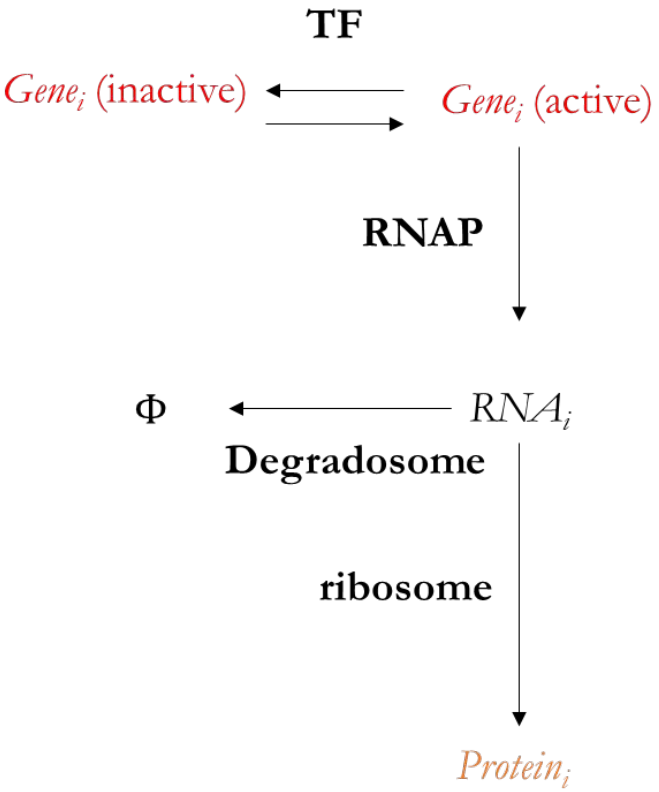


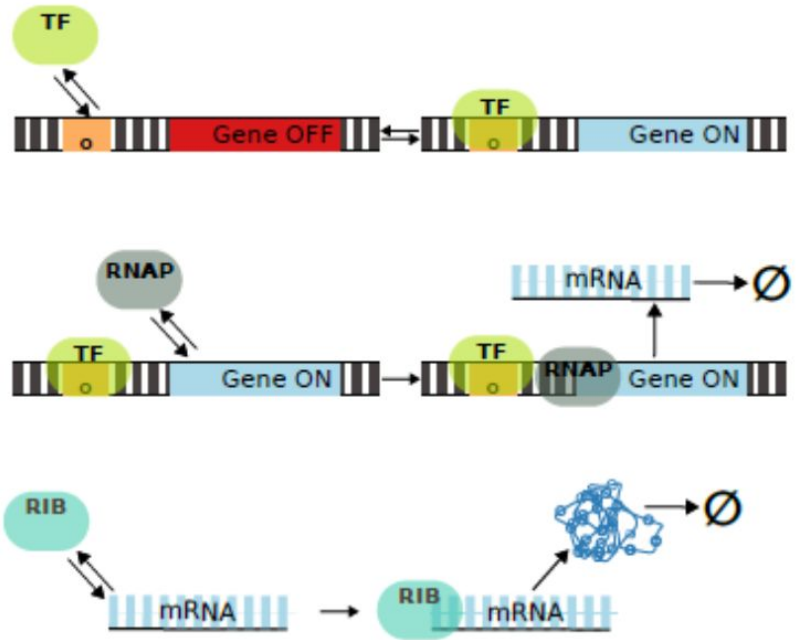
Diffusion limited search



Discrete jumps!

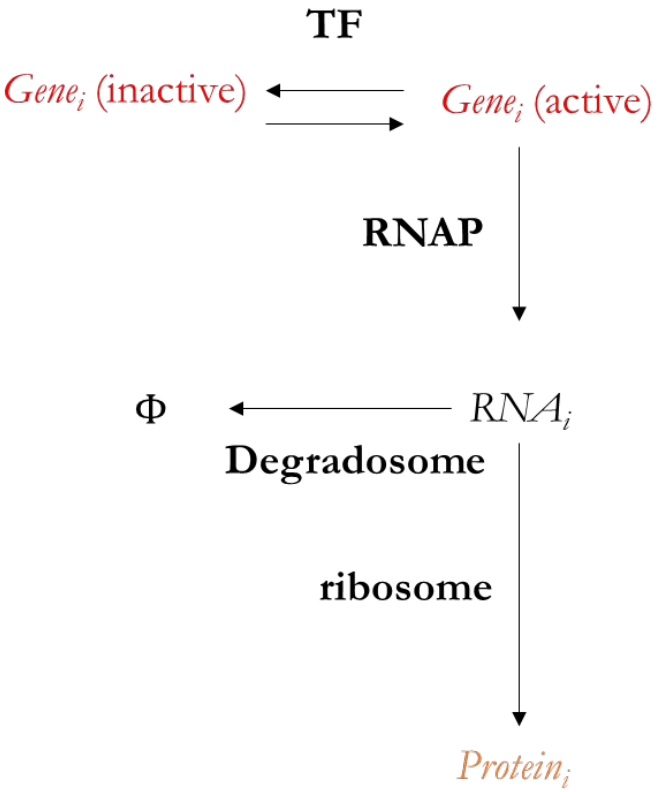
Stochastic steps in gene-expression

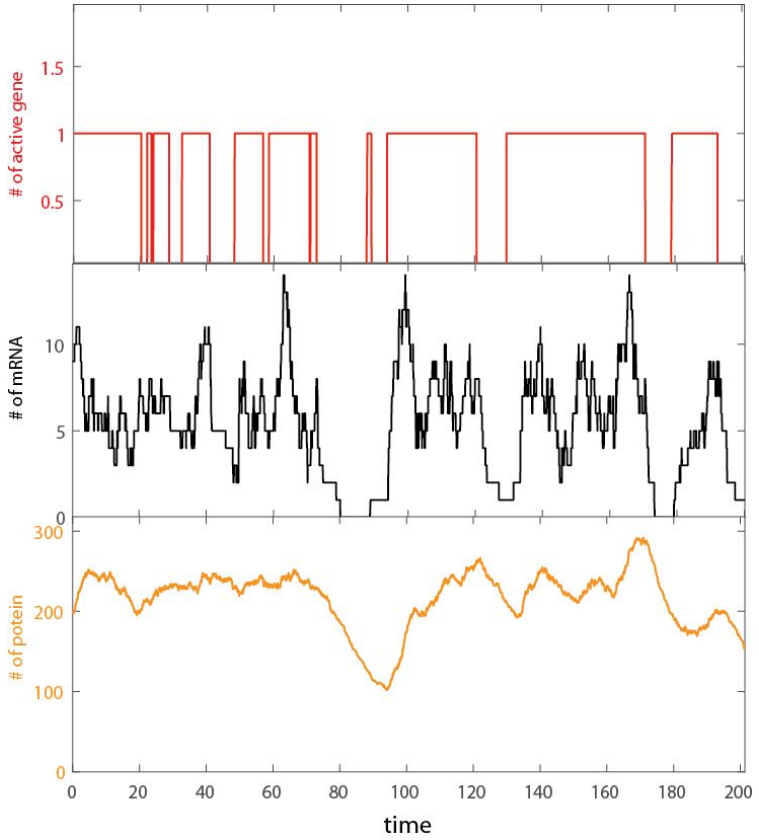




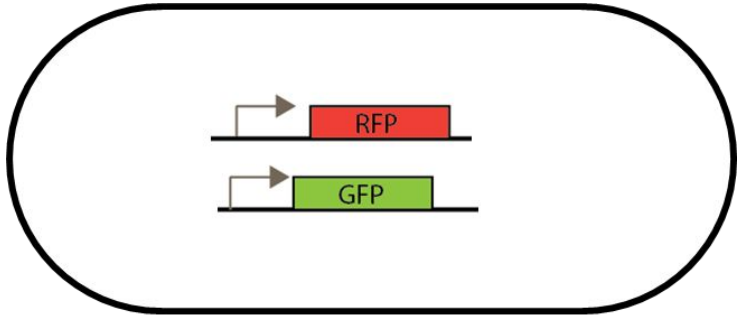
Hardo et al., *Essays in Biochem*. 2011

Stochastic gene-expression



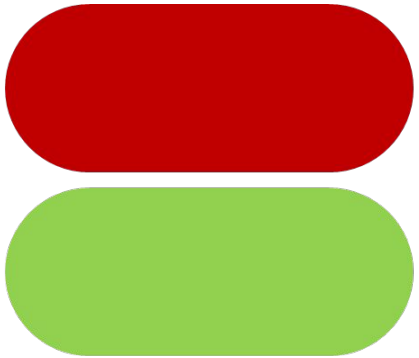


Dual reporter experiment



Dual-reporter system

Elowitz et al., Science 2001

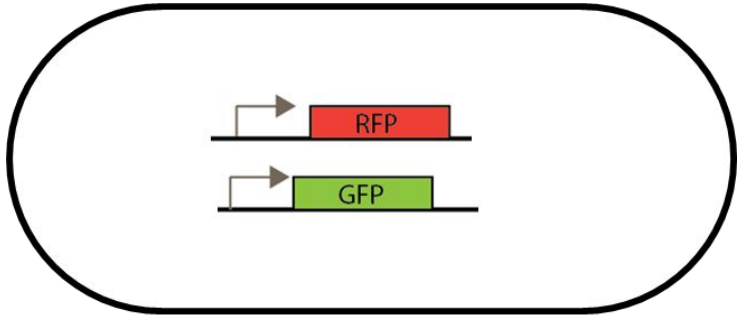


Should have same steady-state level



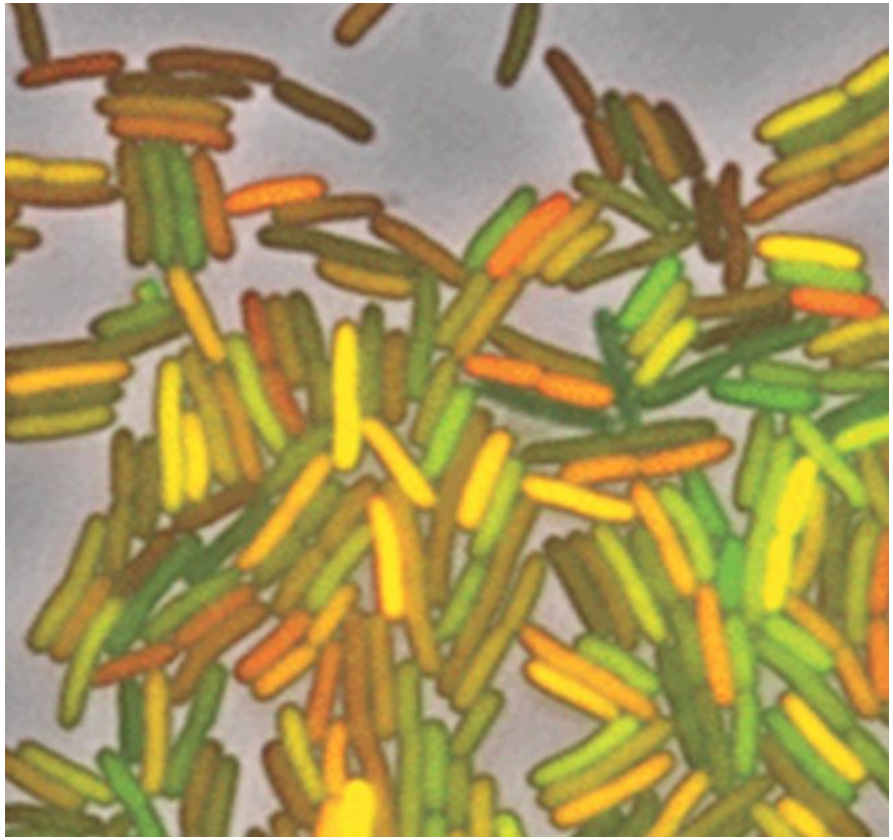
Cells should appear yellow

Dual reporter experiment

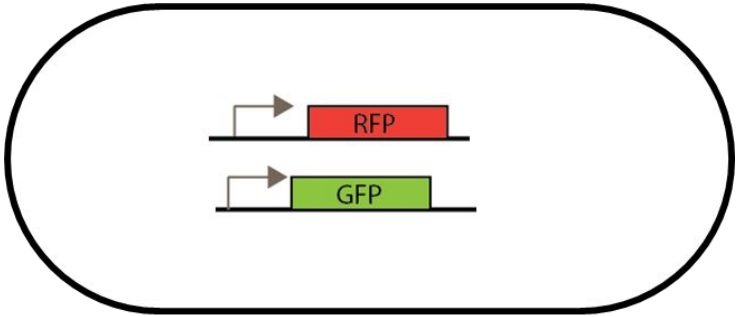


Dual-reporter system

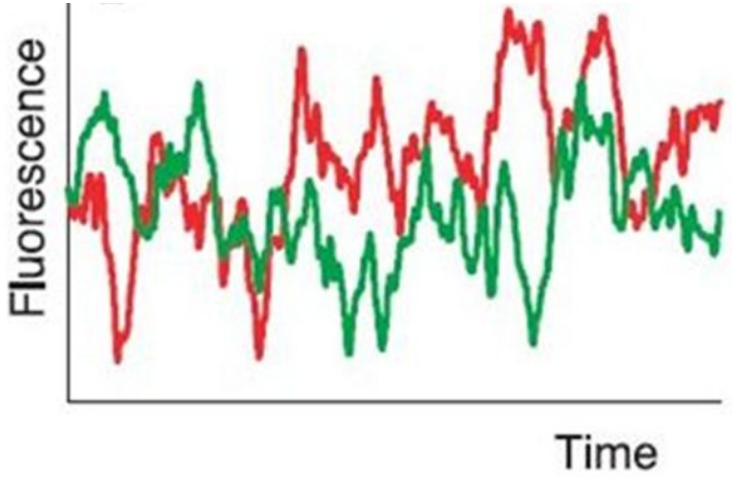
Elowitz et al., Science 2001



Dual reporter experiment

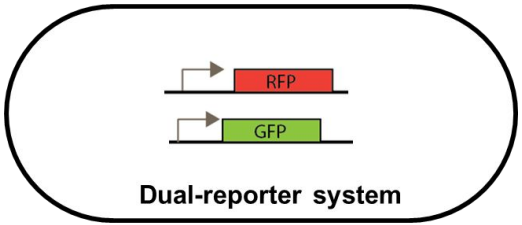


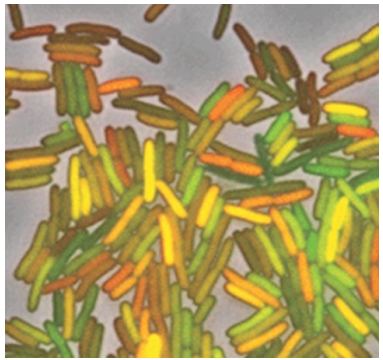
Dual-reporter system

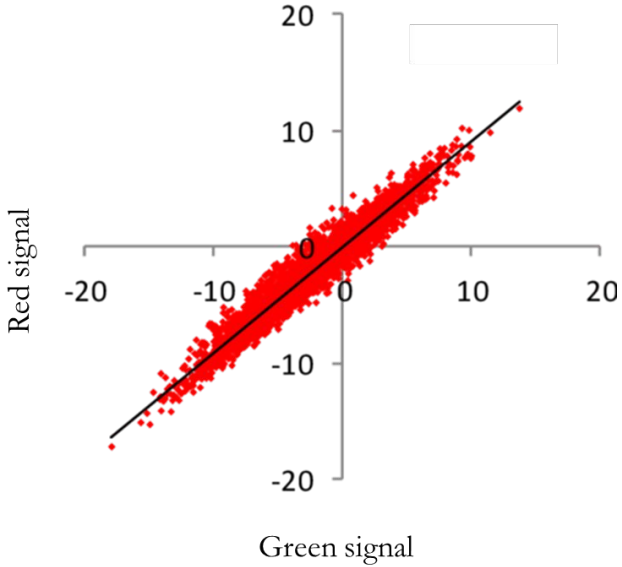


Elowitz et al., Science 2001

Quiz 4



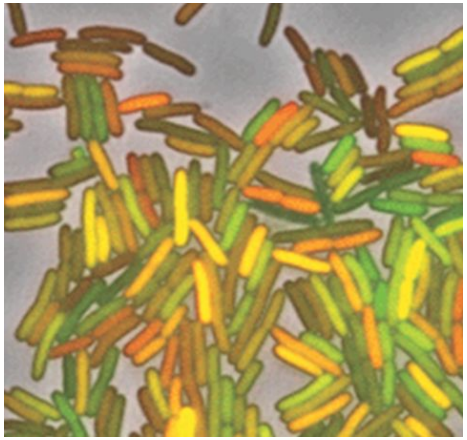


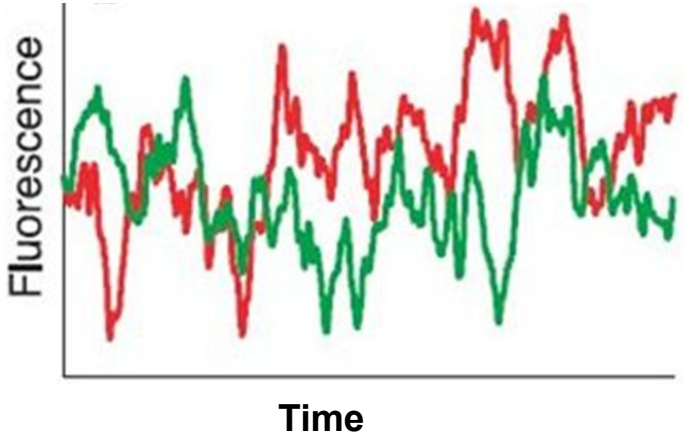


How could you quantify the noise from the environment and the noise that is intrinsic to this system?

Which is higher in the plot on the left?

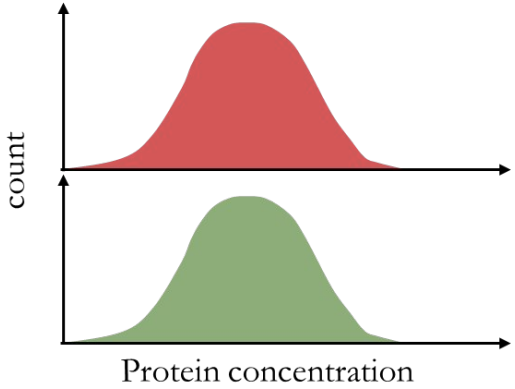
Quantifying stochastic gene-expression

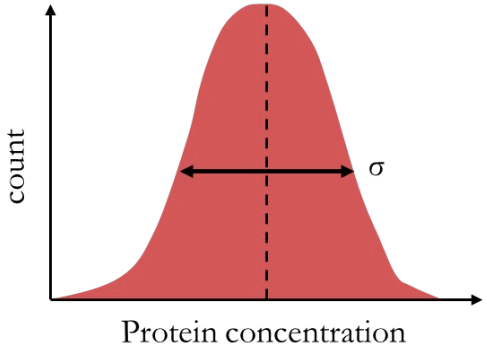




Ergodic process

(only at steady state)





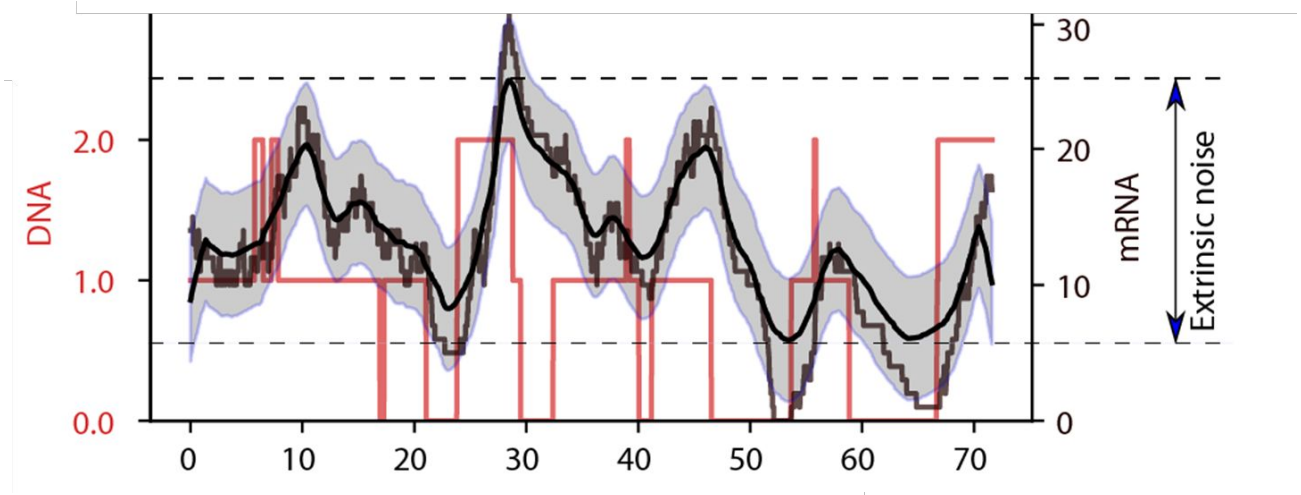
Coefficient of variation

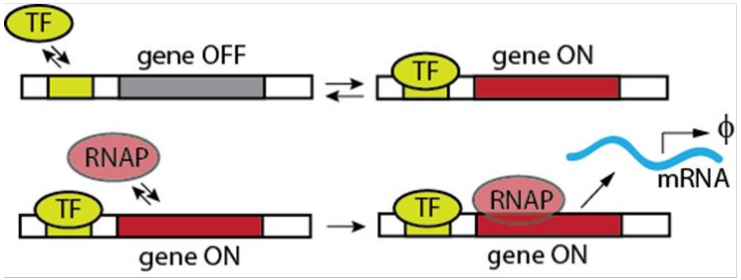
CV $= \sigma/\mu$;

μ = mean amount of protein

σ = standard deviation

Propagation of noise: Extrinsic noise

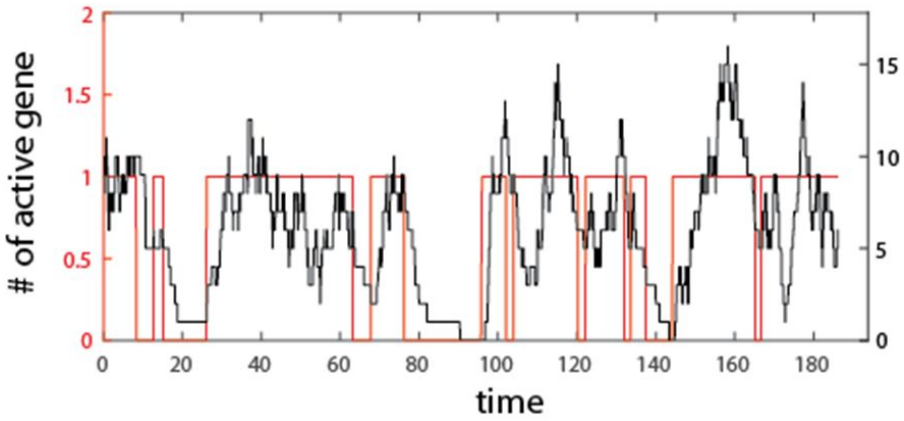


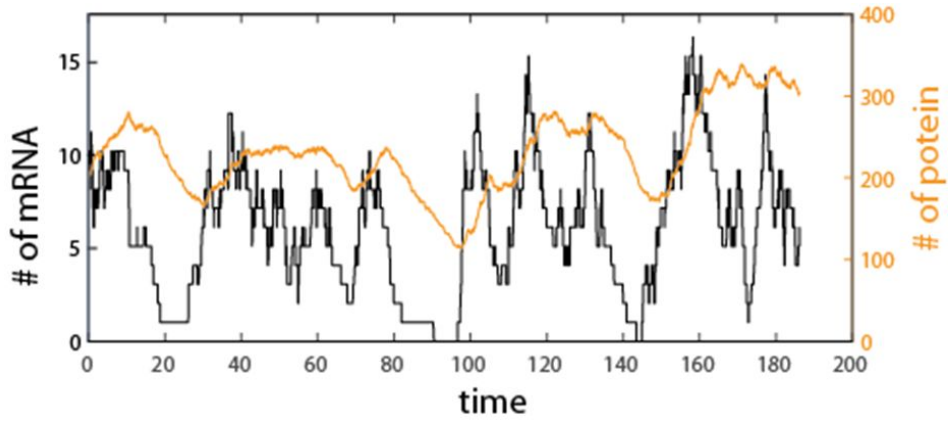


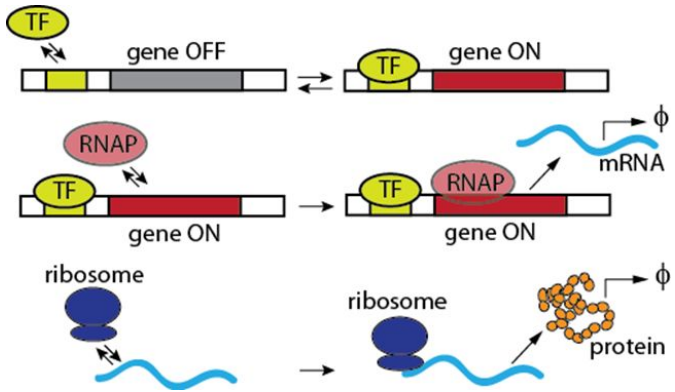
$$\frac{\sigma_g^2}{\langle g \rangle^2} = \frac{1-P_{on}}{\langle g \rangle}$$

$$\frac{\sigma_m^2}{\langle m \rangle^2} = \underbrace{\frac{1}{\langle m \rangle}}_{\text{Intrinsic noise}} + \underbrace{\frac{1-P_{on}}{\langle g \rangle}\frac{\tau_g}{(\tau_g+\tau_m)}}_{\text{Extrinsic noise}}$$

Stochastic gene-expression





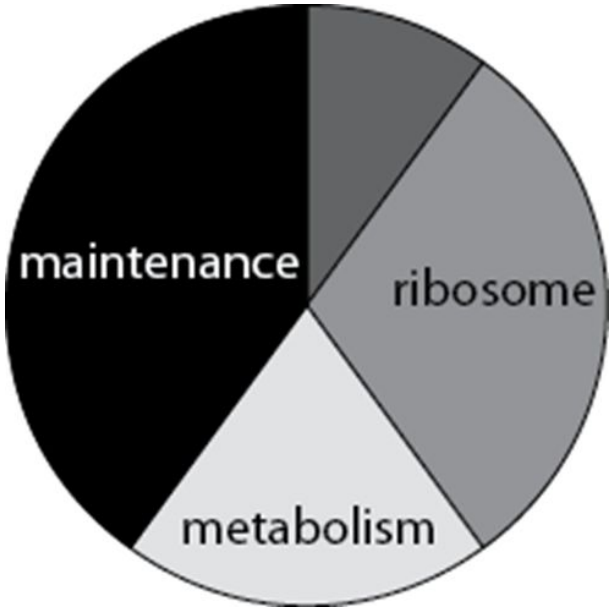


$$\frac{\sigma_g^2}{\langle g\rangle^2} = \frac{1-P_{on}}{\langle g\rangle}$$

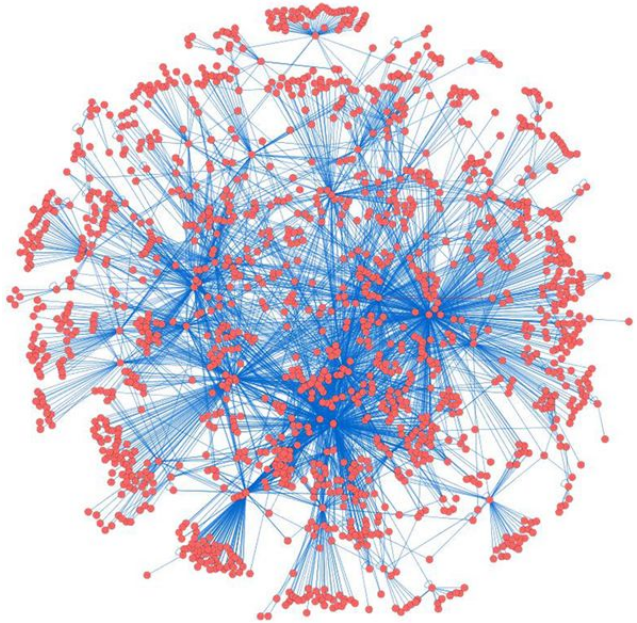
$$\frac{\sigma_m^2}{\langle m\rangle^2} = \frac{1}{\langle m\rangle} + \frac{1-P_{on}}{\langle g\rangle}\frac{\tau_g}{(\tau_g+\tau_m)}$$

$$\frac{\sigma_p^2}{\langle p\rangle^2} = \underbrace{\frac{1}{\langle p\rangle}}_{\text{Intrinsic noise}} + \underbrace{\frac{1}{\langle m\rangle}\frac{\tau_m}{\tau_m+\tau_p} + \frac{1-P_{on}}{\langle g\rangle}\frac{\tau_m}{\tau_m+\tau_p}\frac{\tau_g}{\tau_g+\tau_p}\frac{\tau_g+\tau_p+\frac{\tau_g\tau_p}{\tau_m}}{\tau_g+\tau_m}}_{\text{Extrinsic noise}}$$

Noise from shared network and resources



Cellular resource sharing

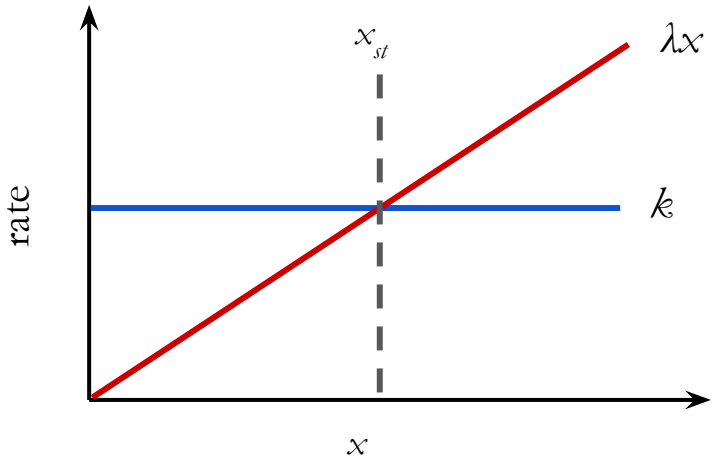


Gene regulatory network of *E. coli*

How can we make our expression system less sensitive to noise in input?

Rate plots for sensitivity analysis

$$\frac{dx}{dt} = k - \lambda x \qquad \text{(Simple regulation)}$$

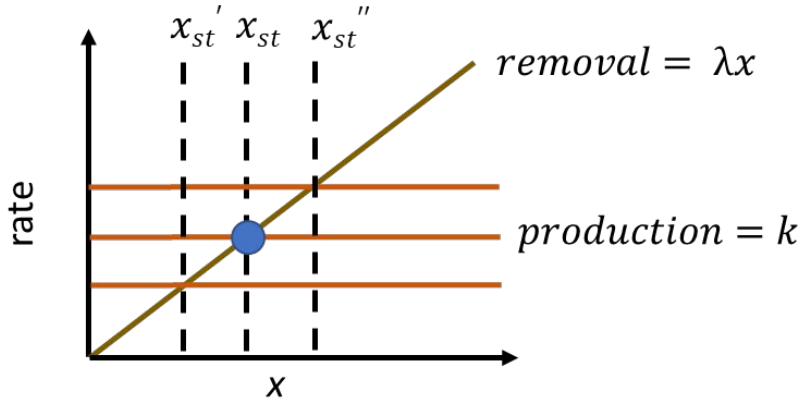


Sensitivity to extrinsic noise: Simple regulation

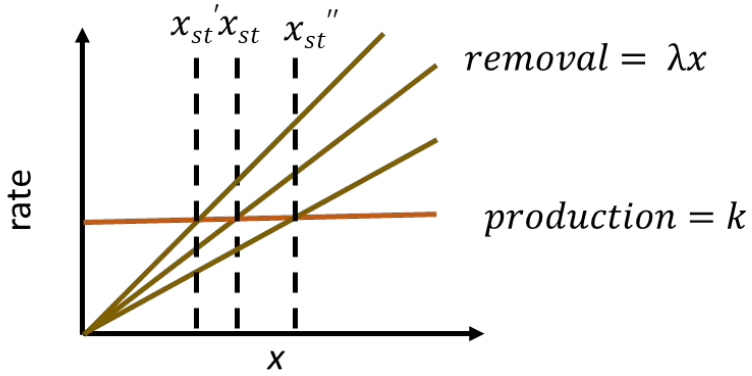
Simple regulation



(No feedback)

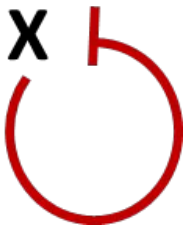


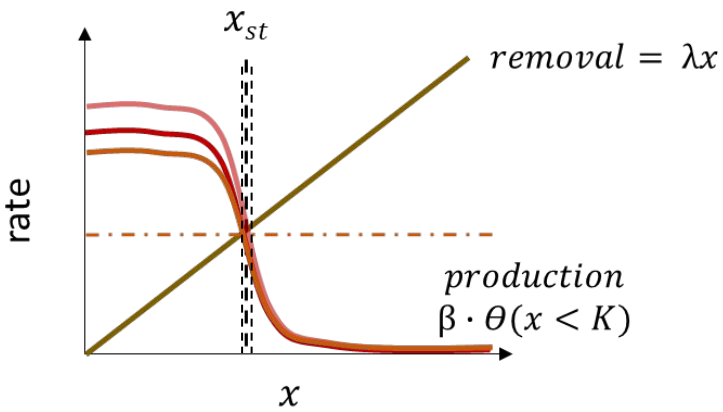
e.g. variability in ribosome/RNAP content

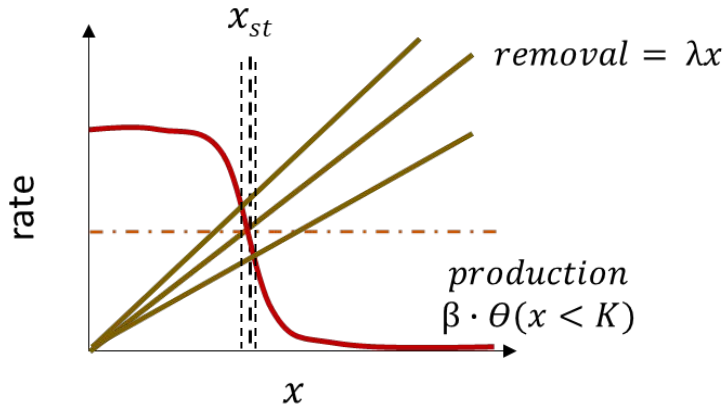


Dilution is more robust, since growth is precise
Degradation can have fluctuation from degradasomes

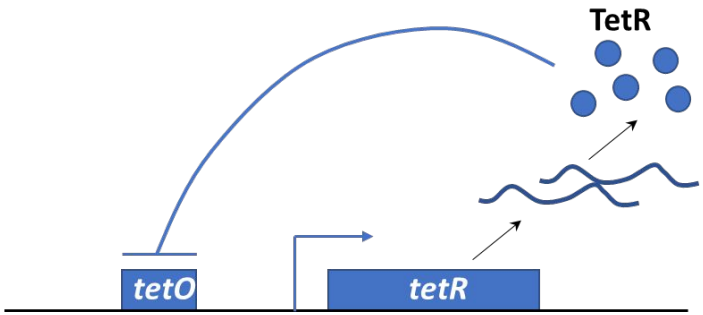
Robustness to extrinsic noise: NAR

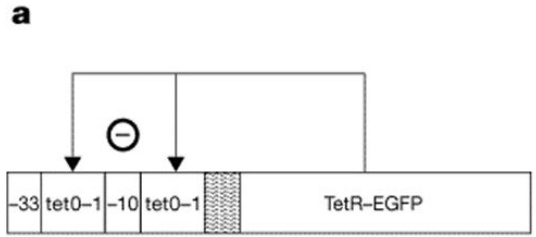


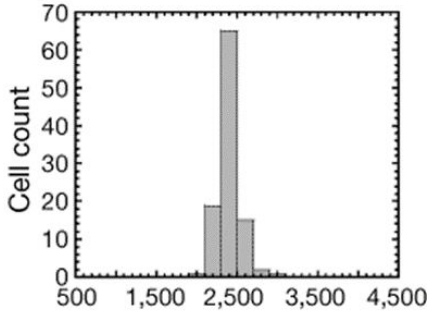


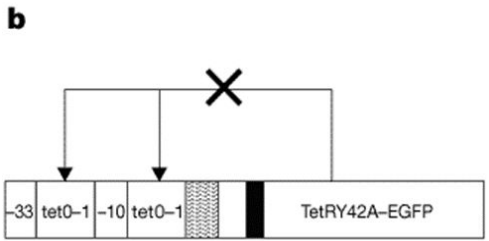


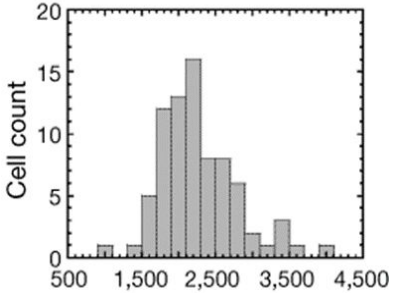
Negative feedback reduces variability

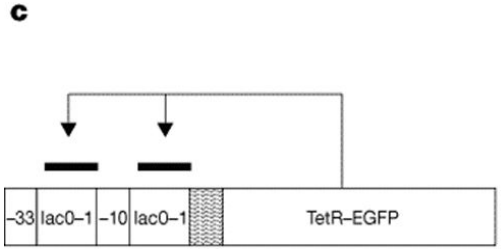


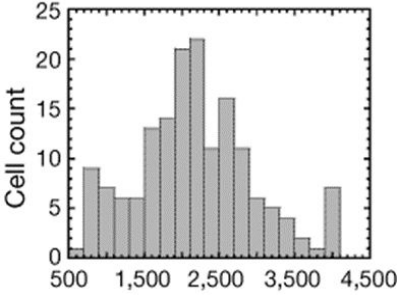






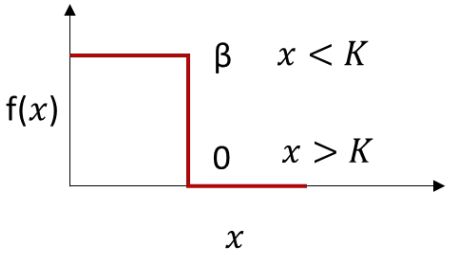


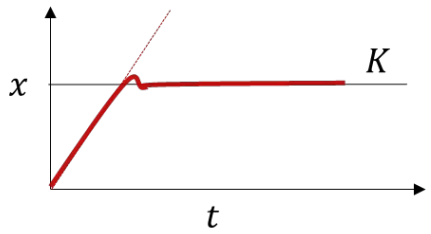


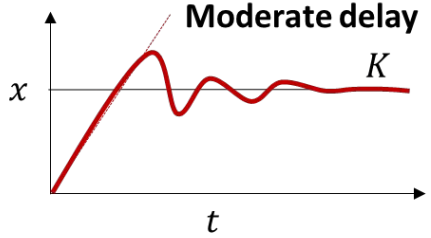


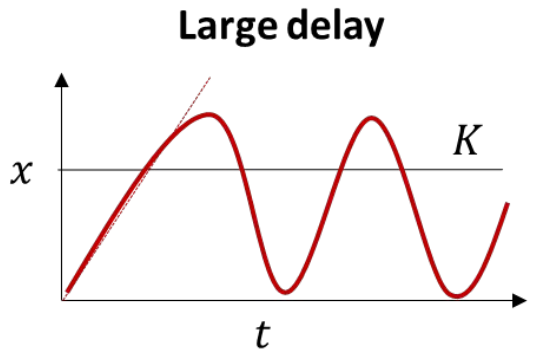
PMID: 10850721

Negative feedback and delay

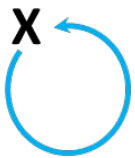


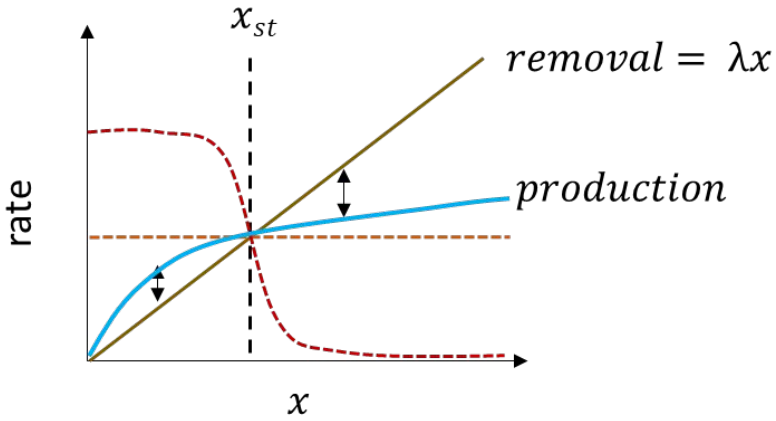


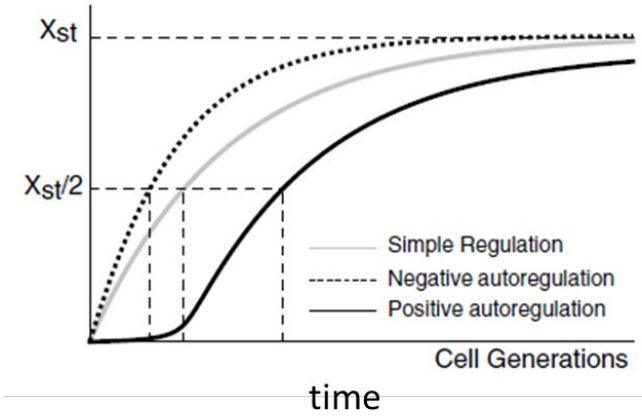




Positive autoregulation (PAR) slows down response time

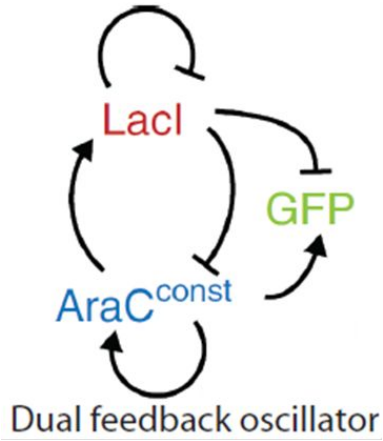




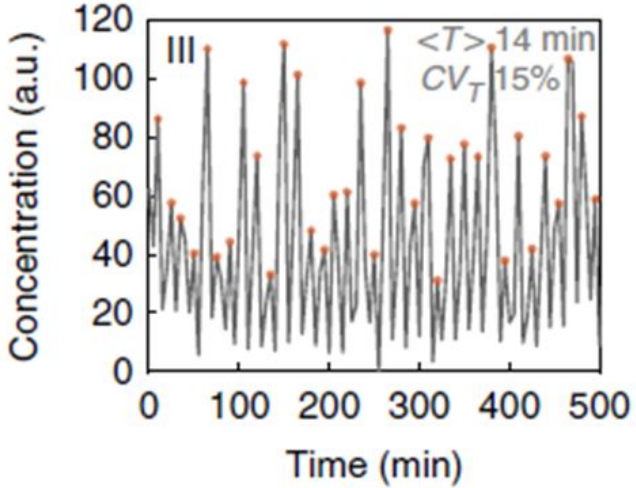


Synthetic genetic oscillators

Delay with additional steps
(+ positive feedback)



Stricker *et al*., 2008

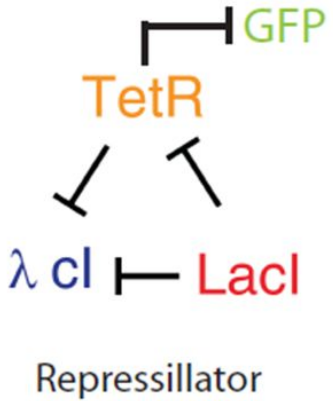


Luro *et al*. 2020

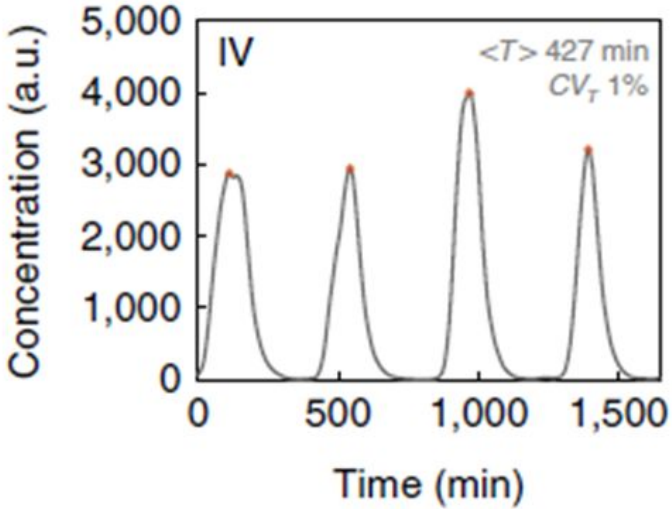
Tunable fast oscillator

Synthetic genetic oscillators

Delay with additional steps

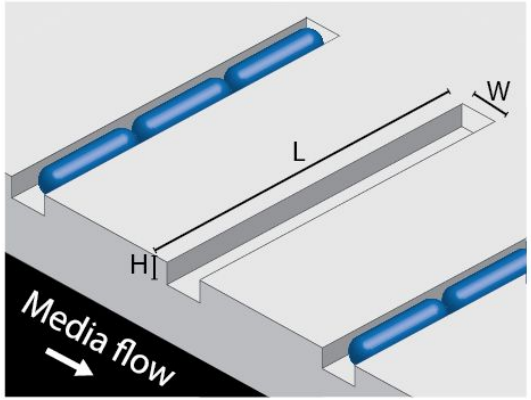


Elowitz *et al*. 2000

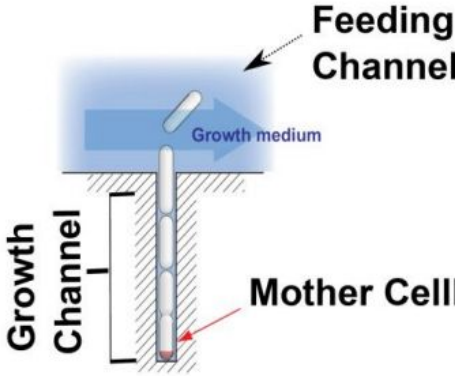


Luro *et al*. 2020

Spatiotemporally uniform growth conditions: Microfluidics



Uniform growth conditions

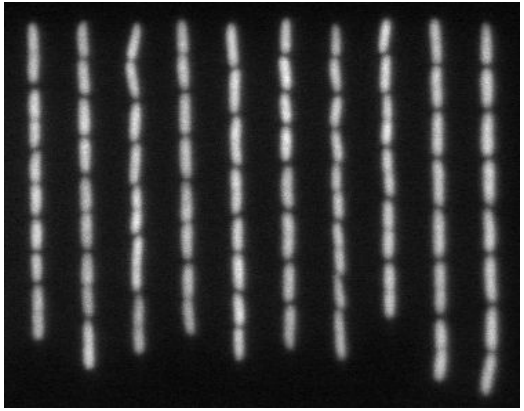


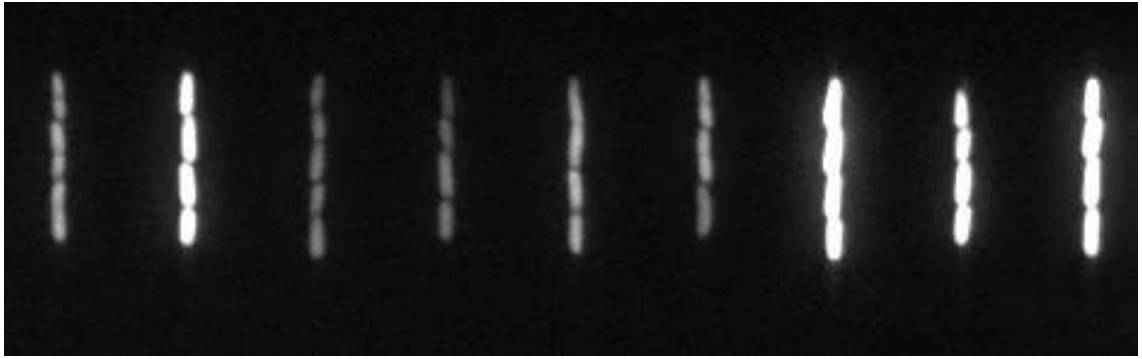
Mother machine:

- Uniform growth conditions across cell trenches
- Steady growth conditions over time

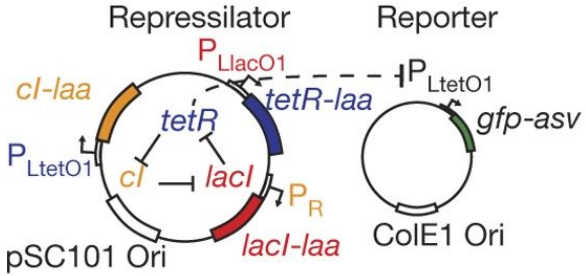
Elowitz Nature 2000

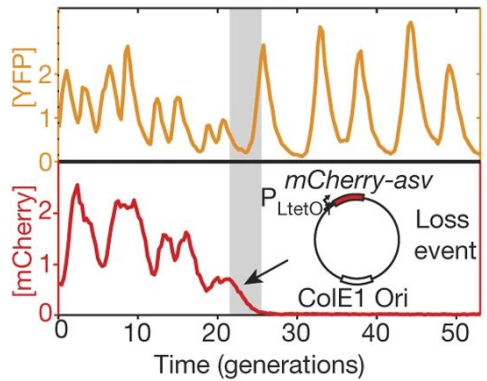
Potvin et al., *Nature* 2016





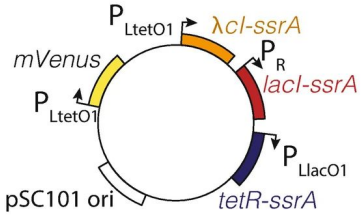
Improving the genetic oscillator

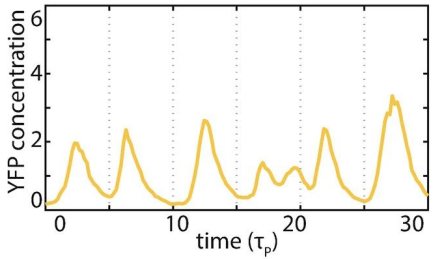




a. amplitude noise

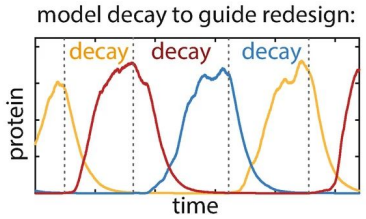
- integrate reporter to remove noise from reporter plasmid

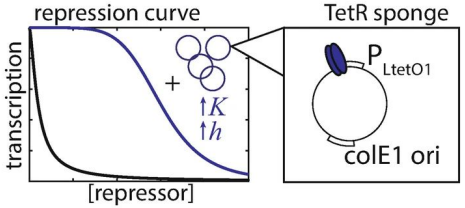




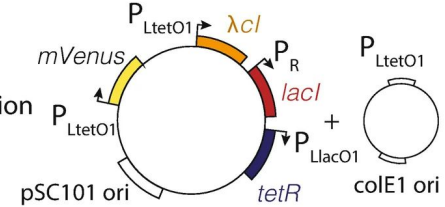
b. noisy decay

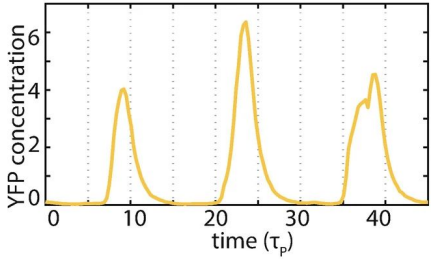
- remove degradation to increase peak protein number





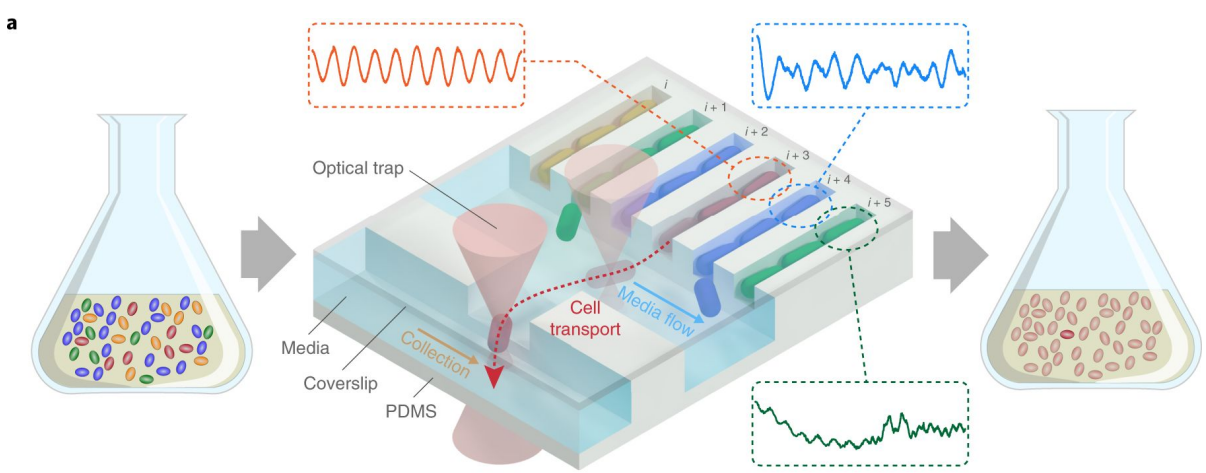
- add titration sponge to increase repression threshold and cooperativity

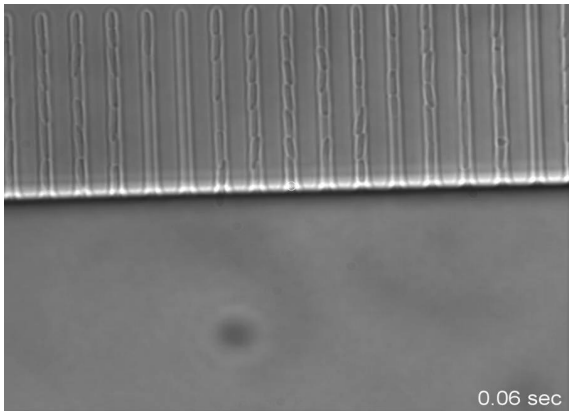




(Potvin et al., *Nature* 2016)

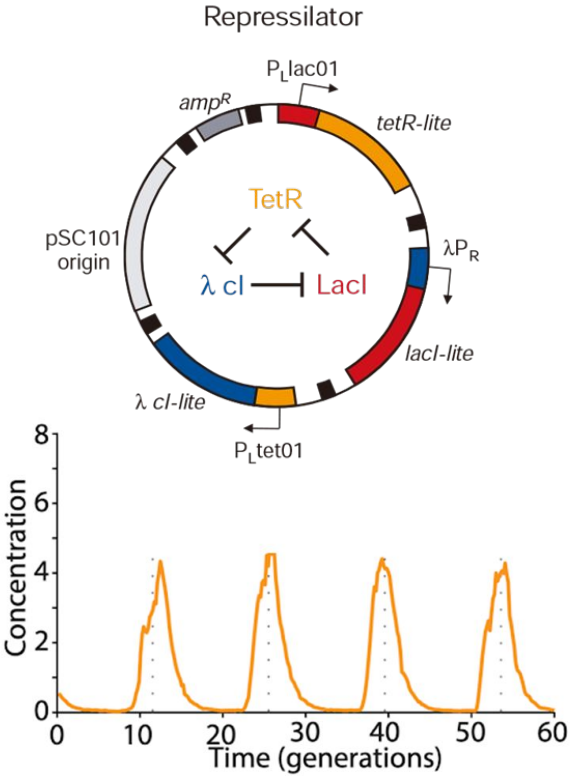
Genetic circuit optimisation: optical tweezers

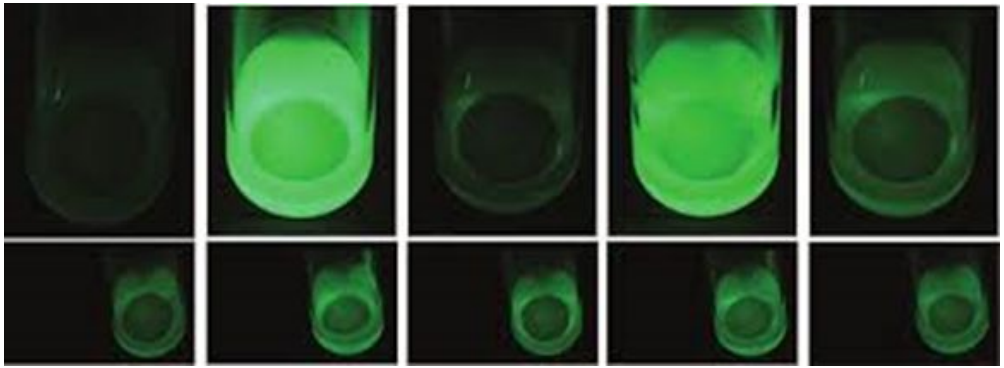




SIFT - Luro et al., *Nature Methods*, 2020

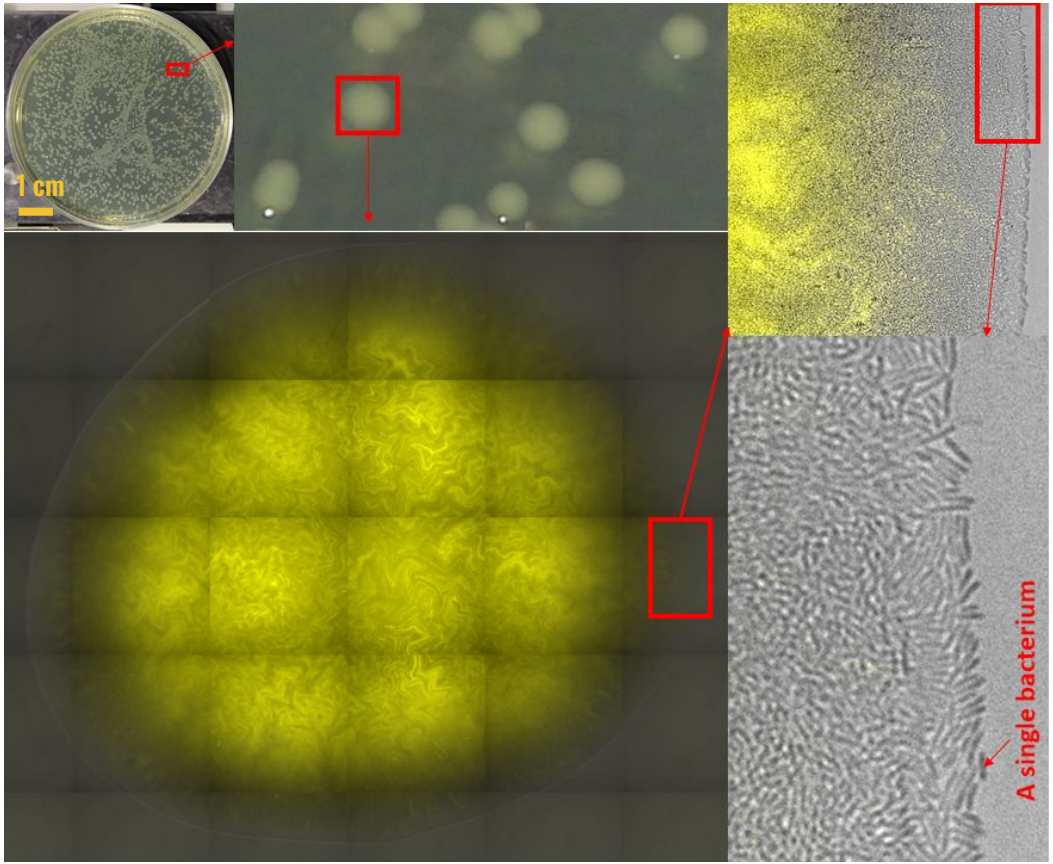
Repressilator as a robust and precise timer

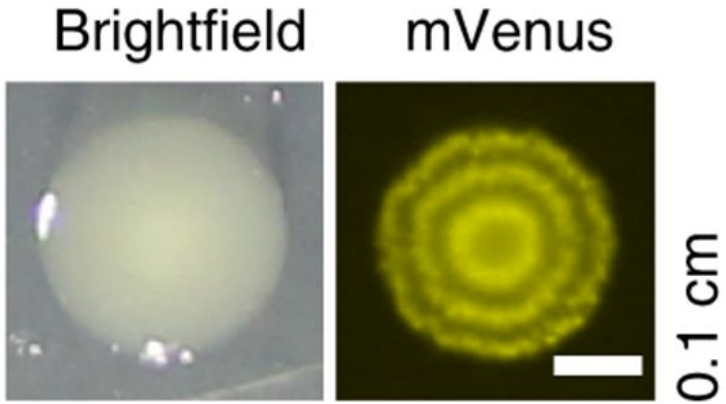




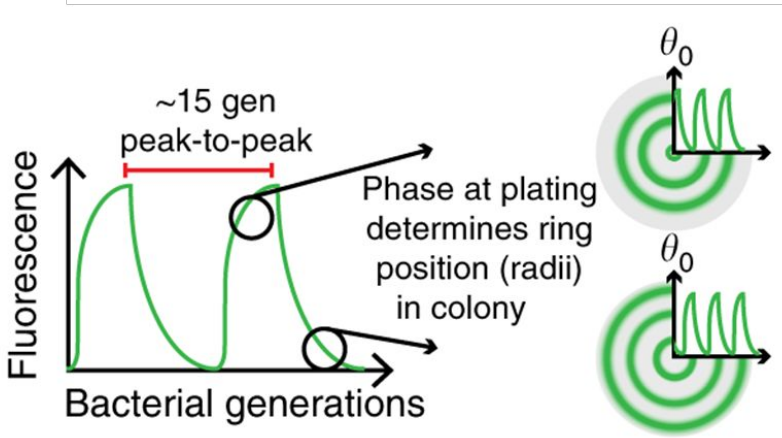
Time

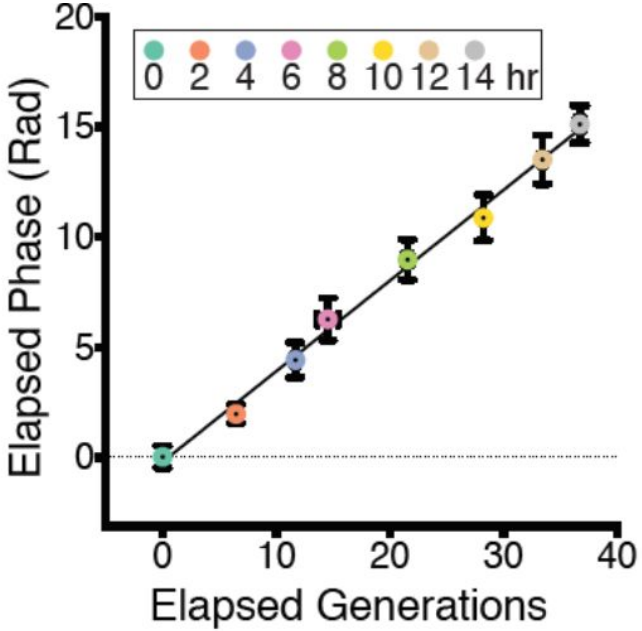
Megacities of microbes



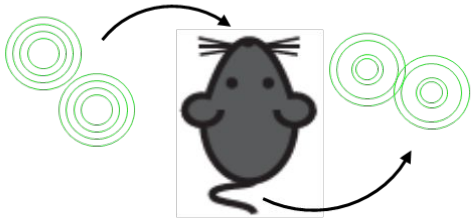


Repressilator as a timer in mammalian gut





Monitoring growth-dynamics in gut



Diagnostics for gut inflammation

Riglar *et al.*, Nat. Comm 2019